

A bicyclic 30-vertex graph with domination root -4

A counterexample to the proposed 33-vertex minimum

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Abstract

We construct a connected simple graph on 30 vertices and 31 edges whose domination polynomial has the simple integer root -4 . This answers negatively the explicit question whether 33 is the minimum possible order. The graph is planar, has treewidth two, and has domination number ten. The root certificate reduces to the rational identity

$$21 - \frac{5}{13} + \frac{256}{13} \left(-\frac{1}{2} - \frac{35}{64} \right) = 0.$$

We prove the reduction, give the complete edge list and polynomial, and provide three mutually consistent exact polynomial certificates. A supported-edge construction gives connected planar examples at every order at least 32, and examples with arbitrarily high multiplicity at -4 . The accompanying dependency-free Python programs reproduce the discovery and verify the certificates. No claim is made that 30 is the true minimum order.

Main exact result. There is a connected planar graph G_{30} with

$$|V| = 30, \quad |E| = 31, \quad \text{tw}(G_{30}) = 2, \quad D(G_{30}, x) = x^{10}(x + 4)Q_{19}(x),$$

where $Q_{19} \in \mathbb{Z}[x]$ and $Q_{19}(-4) = -38\,318\,272 \neq 0$. The graph, not merely an abstract existence argument, is specified below.

1 The question selected and the scope of the answer

For a finite simple graph $G = (V, E)$, a set $S \subseteq V$ is *dominating* when every vertex outside S has a neighbor in S . The domination polynomial is the generating polynomial

$$D(G, x) = \sum_{\substack{S \subseteq V \\ S \text{ dominating}}} x^{|S|} = \sum_{k=0}^{|V|} d(G, k) x^k. \quad (1.1)$$

A domination root is a zero of this polynomial. The coefficients count sets; all evaluations and cancellations in this note are exact.

Alikhani and Griswold [1] gave a 33-vertex graph with domination root -4 , already disproving the earlier conjecture that the only integer domination roots are 0 and -2 . Their Section 4, question 2, asks whether 33 is minimal. We attack that follow-up question, not the already-refuted original conjecture. The present construction shows that 33 is *not* minimal. It also reduces the cyclomatic number from four in their example to two.

The result is an explicit finite certificate with a conventional proof. It is not a claim of a globally smallest graph, a classification of integer domination roots, or a resolution of whether -6 occurs. Literature searches through the date of this note did not locate the same improvement, but cannot establish publication priority. Independent human review remains appropriate before making a novelty claim in a submission.

2 The graph and its elementary invariants

Let $V(G_{30}) = \{0, 1, \dots, 29\}$. Its edge set is the following set of unordered pairs:

$$\begin{aligned}
 E(G_{30}) = \{ & \{0, 1\}, \{1, 2\}, \{2, 3\}, \{3, 0\}, \{1, 4\}, \{2, 4\}, \\
 & \{4, 5\}, \{4, 6\}, \{4, 7\}, \{4, 8\}, \{4, 9\}, \{1, 10\}, \{10, 11\}, \\
 & \{2, 12\}, \{12, 13\}, \{12, 14\}, \{14, 15\}, \{15, 16\}, \\
 & \{14, 17\}, \{17, 18\}, \{14, 19\}, \{19, 20\}, \{14, 21\}, \\
 & \{21, 22\}, \{21, 23\}, \{21, 24\}, \{3, 25\}, \{25, 26\}, \\
 & \{25, 27\}, \{27, 28\}, \{28, 29\} \}.
 \end{aligned}
 \tag{2.1}$$

For machine use, `data/graph30.json` and `data/edges.txt` contain the same edge records as pairs of integers.

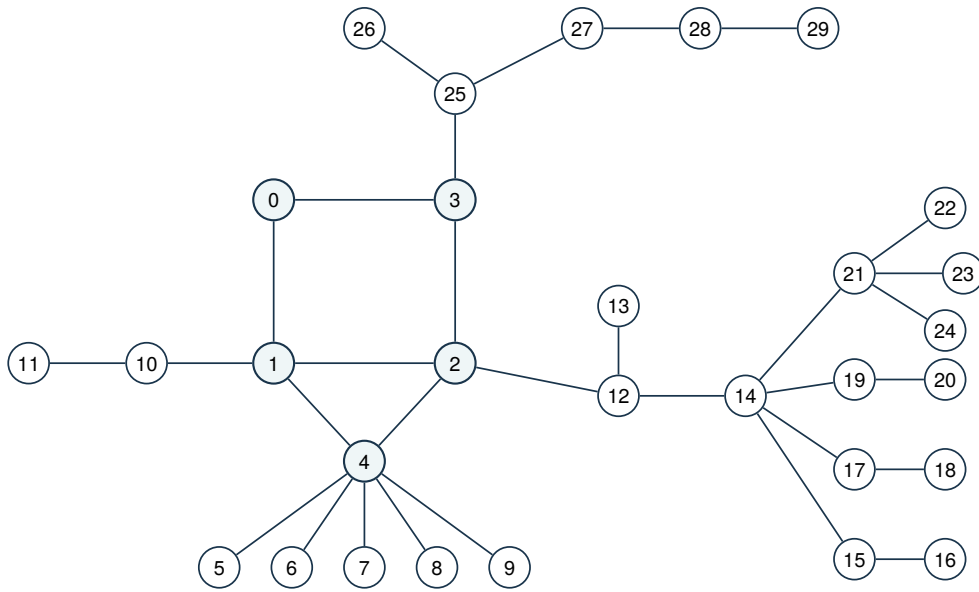


Figure 1: The exact 30-vertex graph. Shaded vertices form its five-vertex 2-core. The displayed drawing is planar.

An equivalent structural description starts with the cycle $0 - 1 - 2 - 3 - 0$. Add a common neighbor 4 of vertices 1, 2, and give 4 five leaves. Attach to vertex 1 a two-vertex rooted path. Attach to vertex 2 the rooted tree on vertices $12, \dots, 24$. Attach to vertex 3 the rooted tree on vertices $25, \dots, 29$. Only the root of each attached tree meets the cycle, by one added edge. Vertex 0 has no attached tree.

Proposition 2.1. *The graph G_{30} is connected and planar, has 30 vertices, 31 edges, 15 leaves, maximum degree seven, cyclomatic number two, treewidth two, and domination number ten.*

Proof. The construction gives connectivity and the counts directly. Removing all pendant trees leaves the cycle together with the vertex 4 and its two edges to 1, 2. This core is planar; attaching trees preserves planarity. For a connected graph the cyclomatic number is $|E| - |V| + 1$, which here is $31 - 30 + 1 = 2$.

For completeness, a tree decomposition of the core has the three bags

$$\{1, 2, 4\}, \quad \{0, 1, 2\}, \quad \{0, 2, 3\},$$

joined in the displayed order. Every core edge lies in a bag, and the bags containing any fixed vertex form a connected subtree. Attach one bag $\{u, v\}$ for each edge in a pendant tree, extending from the bag

containing its parent vertex. The largest bag has size three, so the width is at most two. The graph contains a cycle and is not a forest, so its treewidth is not at most one.

To establish the domination number without the polynomial calculation, consider the nine pairwise disjoint closed neighborhoods of leaves

$$\{4, 5\}, \{10, 11\}, \{12, 13\}, \{15, 16\}, \{17, 18\}, \{19, 20\}, \{21, 22\}, \{25, 26\}, \{28, 29\}.$$

Every dominating set must meet each one. It must also meet $N[0] = \{0, 1, 3\}$, disjoint from all nine. Thus its size is at least ten. The set

$$\{0, 4, 10, 12, 15, 17, 19, 21, 25, 28\}$$

dominates every vertex, proving the matching upper bound. $\square$

The cyclomatic number has an additional interpretation. Oboudi's Theorem 7 [2] excludes integer roots other than 0, -2 for trees and unicyclic graphs. Combining that published result with the construction below gives

$$\min\{|E(G)| - |V(G)| + 1 : G \text{ connected}, D(G, -4) = 0\} = 2. \quad (2.2)$$

The upper bound in (2.2) is proved here; its lower bound uses the cited theorem. This is an exact minimum of *cycle rank*, not an assertion that 30 is a minimum of *order*.

3 Rooted-tree states: the local calculation

The three-state recurrence below is elementary. We include the details so that the root certificate does not rely on any black-box graph polynomial implementation.

Definition 3.1. For a tree T with distinguished root v , let $A_T, B_T, C_T \in \mathbb{Z}[x]$ be the weighted sums of subsets of $V(T)$ with these respective properties:

- (a) every vertex is dominated and v is selected;
- (b) every vertex is dominated and v is not selected;
- (c) every vertex except v is dominated, and v is not dominated.

The weight of a subset S is $x^{|S|}$. In state C_T , the root is necessarily unselected. Set $P_T = A_T + B_T = D(T, x)$.

Write $\mathcal{R}(T_1, \dots, T_s)$ for the rooted tree obtained by adjoining one new root and joining it to the roots of disjoint copies of $T_1, \dots, T_s$. In particular, $L = \mathcal{R}()$ is one vertex.

Lemma 3.2 (Rooted recurrence). For $T = \mathcal{R}(T_1, \dots, T_s)$,

$$A_T = x \prod_{j=1}^s (A_{T_j} + B_{T_j} + C_{T_j}), \quad (3.1)$$

$$B_T = \prod_{j=1}^s (A_{T_j} + B_{T_j}) - \prod_{j=1}^s B_{T_j}, \quad (3.2)$$

$$C_T = \prod_{j=1}^s B_{T_j}. \quad (3.3)$$

Empty products are one, so $(A_L, B_L, C_L) = (x, 0, 1)$.

Proof. If the new root is selected, it contributes x , and each child root may be in any of the three states: a formerly undominated child root is now dominated by its selected parent. This proves (3.1). If the new root is unselected, each child subtree must internally dominate its own root, allowing only A and B . Among these choices, precisely those in which every child is in state B leave the new root undominated. Subtracting that class gives (3.2), and retaining it gives (3.3). All decompositions preserve the number of selected vertices, proving the polynomial identities. $\square$

If a rooted tree has a leaf immediately adjacent to its root, then $C_T = 0$ as a polynomial: such a leaf contributes the factor $B_L = 0$ to (3.3). Combinatorially, an undominated root would force its private leaf to be both unselected and undominated.

For our graph define

$$\begin{aligned} E &= \mathcal{R}(L), & J &= \mathcal{R}(E), & S_3 &= \mathcal{R}(L, L, L), \\ W &= \mathcal{R}(E, E, E, S_3), & T &= \mathcal{R}(L, W), & K &= \mathcal{R}(L, J), \\ S_5 &= \mathcal{R}(L, L, L, L, L). \end{aligned}$$

The four external gadgets are E, T, K, S_5 , of orders 2, 13, 5, 6. Their roots are respectively the vertices 10, 12, 25, 4. The first three attach to one cycle vertex; the last attaches to both 1, 2. Each of the four has $C = 0$ identically.

Tree	Order	$A(-4)$	$B(-4)$	$C(-4)$	$B/(A+B)$ at -4
L	1	-4	0	1	—
E	2	12	-4	0	-1/2
J	3	-32	12	-4	—
S_3	4	108	-64	0	-16/11
W	11	-90 112	18 432	4 096	—
T	13	-811 008	286 720	0	-35/64
K	5	-288	80	0	-5/13
S_5	6	972	-1 024	0	256/13

Table 1: Exact rooted states. Dashes suppress ratios not used as zero- C attachments. Every entry follows from Lemma 3.2.

Here are the larger arithmetic steps, to make the table checkable by hand. For E , $A + B = 8$; for S_3 , $A + B = 44$. Thus

$$\begin{aligned} A_W &= -4 \cdot 8^3 \cdot 44 = -90\,112, \\ B_W &= 8^3 \cdot 44 - (-4)^3(-64) = 18\,432, & C_W &= 4\,096. \end{aligned}$$

Therefore $A_W + B_W + C_W = -67\,584$, while $A_W + B_W = -71\,680$. Adding the new root and its leaf gives

$$\begin{aligned} A_T &= -4(-3)(-67\,584) = -811\,008, \\ B_T &= (-4)(-71\,680) = 286\,720, & C_T &= 0. \end{aligned}$$

Likewise $(A_J, B_J, C_J) = (-32, 12, -4)$, and hence $A_K = -4(-3)(-24) = -288$, $B_K = (-4)(-20) = 80$. Finally $A_{S_5} = -4(-3)^5 = 972$ and $B_{S_5} = (-4)^5 = -1024$.

4 A four-cycle identity and the root proof

We now prove the identity that turns the table into a root certificate. Although the search used weighted states at one value of x , all four gadgets in the final graph satisfy the stronger polynomial condition $C = 0$.

Proposition 4.1 (One hub and one undecorated corner). *Start with C_4 on $0, 1, 2, 3$. Attach zero- C rooted gadgets F_1, F_2, F_3 to vertices $1, 2, 3$, respectively, and attach a zero- C rooted gadget H by joining its root to both $1, 2$. No gadget is attached to 0 . Write*

$$P_i = A_i + B_i, \quad P_H = A_H + B_H, \quad a = \frac{B_1}{P_1}, \quad b = \frac{B_2}{P_2}, \quad c = \frac{B_3}{P_3}, \quad r = \frac{B_H}{P_H}, \quad \mathcal{P} = P_1 P_2 P_3 P_H.$$

As an identity of rational functions (and after clearing denominators, of polynomials), the resulting graph G satisfies

$$D(G, x) = x\mathcal{P}[x^3 + 4x^2 + 6x + 3 - c - r(a + b)]. \quad (4.1)$$

Proof. First sum over configurations that dominate all gadget vertices but impose no domination requirement on the four cycle vertices. A root cannot be in state C , so its total internal weight is P , independently of which adjacent cycle vertices are selected. The unrestricted total is consequently $\mathcal{P}(1+x)^4$.

Apply inclusion–exclusion to the events that specified cycle vertices are undominated. For $U \subseteq \{0, 1, 2, 3\}$, this forces every cycle vertex in $N_{C_4}[U]$ to be unselected. The remaining cycle vertices contribute $(1+x)^{4-|N_{C_4}[U]|}$. Every attached root meeting a vertex of U must also be unselected, replacing its factor P by B . In particular, the shared root contributes r once whenever $U \cap \{1, 2\} \neq \emptyset$, not twice. With $u_0 = 1, u_1 = a, u_2 = b, u_3 = c$, this gives

$$\frac{D(G, x)}{\mathcal{P}} = \sum_{U \subseteq \{0, 1, 2, 3\}} (-1)^{|U|} (1+x)^{4-|N_{C_4}[U]|} \prod_{i \in U} u_i r^{\mathbf{1}_{U \cap \{1, 2\} \neq \emptyset}}. \quad (4.2)$$

Put $t = 1 + x$. The empty-set contribution is t^4 , and singleton contributions sum to $-t[1 + c + r(a + b)]$. Any two distinct cycle vertices have closed-neighborhood union equal to the entire cycle. The contributions by cardinality two, three and four are therefore

$$\begin{aligned} & c + r(a + b + ab + ac + bc), \\ & -r(ab + ac + bc + abc), \\ & rabc, \end{aligned}$$

whose sum is $c + r(a + b)$. Thus (4.2) becomes

$$t^4 - t + (1 - t)[c + r(a + b)] = x[x^3 + 4x^2 + 6x + 3 - c - r(a + b)],$$

proving the assertion. Clearing denominators also proves the identity at values where a displayed ratio is undefined. $\square$

Theorem 4.2 (The counterexample). *The graph specified by (2.1) satisfies $D(G_{30}, -4) = 0$. Consequently, the answer to the 33-vertex minimality question is no.*

Proof. Apply Proposition 4.1 with $F_1 = E, F_2 = T, F_3 = K, H = S_5$. At $x = -4$,

$$(P_1, P_2, P_3, P_H) = (8, -524\,288, -208, -52), \quad \mathcal{P} = -45\,365\,592\,064 \neq 0.$$

Table 1 gives

$$a = -\frac{1}{2}, \quad b = -\frac{35}{64}, \quad c = -\frac{5}{13}, \quad r = \frac{256}{13}.$$

At -4 , the bracket in (4.1) is $-21 - c - r(a + b)$. But

$$a + b = -\frac{67}{64}, \quad r(a + b) = -\frac{268}{13}, \quad 21 + c = \frac{268}{13}.$$

The bracket is exactly zero. Equivalently,

$$\frac{D(G_{30}, -4)}{4\mathcal{P}} = 21 - \frac{5}{13} + \frac{256}{13} \left(-\frac{1}{2} - \frac{35}{64} \right) = 0. \quad (4.3)$$

This is an equality in $\mathbb{Q}$, not a numerical approximation. $\square$

Remark 4.3. The proof uses only the explicit graph and the elementary state and inclusion–exclusion identities proved above. It does not assume the correctness of a numerical factorization, any search program, or any particular normalization of the transfer matrices in earlier work.

5 The full polynomial and independent certificates

The nonzero coefficients are listed in Table 2. All coefficients below degree ten vanish. The last coefficient is one, as required because the whole vertex set is the unique set of size 30.

k	$d(G, k)$	k	$d(G, k)$	k	$d(G, k)$
10	256	17	4 526 996	24	342 684
11	4 192	18	5 803 019	25	99 255
12	32 576	19	6 049 225	26	22 114
13	159 418	20	5 150 665	27	3 651
14	550 047	21	3 583 938	28	420
15	1 420 449	22	2 031 824	29	30
16	2 845 950	23	932 405	30	1

Table 2: The complete nonzero coefficient list of $D(G_{30}, x)$.

Exact division gives

$$D(G_{30}, x) = x^{10}(x + 4)Q_{19}(x), \quad (5.1)$$

where

$$\begin{aligned} Q_{19}(x) = & x^{19} + 26x^{18} + 316x^{17} + 2387x^{16} + 12566x^{15} \\ & + 48991x^{14} + 146720x^{13} + 345525x^{12} + 649724x^{11} \\ & + 985042x^{10} + 1210497x^9 + 1207237x^8 + 974071x^7 \\ & + 630712x^6 + 323102x^5 + 128041x^4 + 37883x^3 \\ & + 7886x^2 + 1032x + 64. \end{aligned} \quad (5.2)$$

No irreducibility assertion is needed. Substitution yields

$$Q_{19}(-4) = -38\,318\,272, \quad D'(G_{30}, -4) = (-4)^{10}Q_{19}(-4) = -40\,179\,620\,380\,672 \neq 0. \quad (5.3)$$

This proves that -4 is a simple root.

5.1 Certificate A: expansion of the four-cycle identity

Proposition 4.1 has the denominator-free form

$$\begin{aligned} D(G, x) = & x[(x^3 + 4x^2 + 6x + 3)P_1P_2P_3P_H \\ & - B_HP_3(B_1P_2 + B_2P_1) - B_3P_1P_2P_H]. \end{aligned} \quad (5.4)$$

The following compact gadget polynomials, together with (5.4), are one symbolic certificate for every coefficient. Let

$$p = x(x + 2), \quad s = x(1 + x)^3 + x^3.$$

Then

$$\begin{aligned}
 P_1 &= p, & B_1 &= x, \\
 P_H &= x(1+x)^5 + x^5, & B_H &= x^5, \\
 P_3 &= x^2(x^3 + 5x^2 + 8x + 3), & B_3 &= x^2(x^2 + 3x + 1), \\
 P_2 &= x(x+1)(x+2)p^3s - x^7, & B_2 &= x(x+1)p^3s - x^7.
 \end{aligned} \tag{5.5}$$

These follow directly from Lemma 3.2. Expanding (5.4) with (5.5) gives Table 2 and (5.1).

5.2 Certificate B: stripping trees and enumerating the 2-core

Remove vertices of current degree zero or one successively. At a removed vertex, compute its three rooted polynomial states and absorb them into its surviving neighbor. The remaining 2-core here has five vertices: 0, 1, 2, 3, 4.

At each remaining core vertex v , let

$$T_v = \prod_j (A_j + B_j + C_j), \quad U_v = \prod_j (A_j + B_j), \quad V_v = \prod_j B_j,$$

where the product runs over its absorbed child trees. For a selected core subset S , the local contribution of v is

$$\begin{cases} xT_v, & v \in S, \\ U_v, & v \notin S \text{ and } N_{\text{core}}(v) \cap S \neq \emptyset, \\ U_v - V_v, & v \notin S \text{ and } N_{\text{core}}(v) \cap S = \emptyset. \end{cases} \tag{5.6}$$

Multiplying local factors and summing over all $2^5 = 32$ subsets gives the full polynomial. Formula (5.6) is exactly the distinction between root selected, root dominated by a core neighbor, and root needing domination from a child. Thus it counts each dominating set once. This calculation does not use the special four-cycle identity.

5.3 Certificate C: an independent enumeration over non-leaves

The following identity avoids rooted states and 2-core stripping altogether.

Proposition 5.1 (Leaf-subset certificate). *Let G be a finite graph with no component isomorphic to K_2 . Let L be its set of degree-one vertices and $C = V \setminus L$. For $v \in C$, set $t_v = |N(v) \cap L|$. Call $S \subseteq C$ admissible when every $v \in C$ with $t_v = 0$ is dominated by S in $G[C]$. Then*

$$D(G, x) = \sum_{S \text{ admissible}} x^{|S| + \sum_{v \in C \setminus S} t_v} (1+x)^{\sum_{v \in S} t_v}. \tag{5.7}$$

Proof. Fix the selected non-leaves S . If a non-leaf v is selected, its adjacent leaves may each be selected or not, independently, giving $(1+x)^{t_v}$. If v is not selected, every one of those leaves must be selected, contributing x^{t_v} . When $t_v > 0$, these forced leaves also dominate v . When $t_v = 0$, the only way to dominate an unselected v is through the selected non-leaves; this is precisely admissibility. The exclusion of isolated-edge components ensures that every leaf has a non-leaf neighbor. These disjoint cases exhaust all dominating sets and prove (5.7). $\square$

For G_{30} , $|L| = |C| = 15$, so the independent computation tests $2^{15} = 32768$ non-leaf subsets. Exactly 23065 are admissible. Grouping the summands in (5.7) by their two exponents gives 118 terms $m_{a,b}x^a(1+x)^b$. The file `data/leaf_certificate.csv` records all triples $(a, b, m_{a,b})$. Binomial expansion reproduces Table 2, and direct substitution in that 118-term sum gives zero at -4 .

6 Connected families and arbitrary multiplicity

There is a simple way to preserve the root without leaving connected graphs. The underlying supported-edge factorization is elementary and related to the pendant-star products in Oboudi [2]; it is not claimed as a new general principle here.

Call a vertex a *support vertex* when it has a neighbor of degree one, called a private leaf.

Lemma 6.1 (Edges between support vertices). *Suppose distinct vertices u, v each have a private leaf outside $\{u, v\}$, and these leaves remain private after the edge uv is added or deleted. Toggling uv does not change the family of dominating vertex subsets, hence does not change the domination polynomial.*

Proof. A dominating set must select either a private leaf of u or u itself, because it must dominate that leaf. Either choice already dominates u , independently of the edge uv . The same holds for v . Every other vertex has exactly the same neighbors before and after the change. Thus a vertex subset dominates one graph exactly when it dominates the other. $\square$

For disjoint graphs, independently choosing dominating sets proves

$$D(G \dot{\cup} H, x) = D(G, x)D(H, x). \quad (6.1)$$

Combining this with the lemma connects factors without changing the product.

Theorem 6.2 (A bounded-width connected family). *For each integer $t \geq 0$, there is a connected planar graph G_t with $30 + 2t$ vertices, cyclomatic number two, treewidth two, maximum degree seven, and*

$$D(G_t, x) = [x(x + 2)]^t D(G_{30}, x). \quad (6.2)$$

In particular, -4 is a simple root of every $D(G_t, x)$.

Proof. For $t = 0$, use G_{30} . For $t \geq 1$, take G_{30} and t disjoint edges $a_i b_i$. Retain each b_i as a private leaf. Join vertex 10 to a_1 , and join a_i to a_{i+1} for $1 \leq i < t$. Vertex 10 retains its leaf 11. Each added connecting edge joins support vertices and therefore changes no dominating sets. Since $D(K_2, x) = 2x + x^2 = x(x + 2)$, equations (6.1) and (6.2) follow.

The new part is a pendant tree, so planarity, treewidth two, and cycle rank two are preserved. There are $2t$ new vertices and $2t$ new edges. Vertex 10 acquires at most one new incident edge; every a_i has degree at most three. The old degree-seven vertex 4 remains, so the maximum degree is still seven. Finally $(-4)(-4 + 2) = 8 \neq 0$, so multiplication by $[x(x + 2)]^t$ does not alter the multiplicity at -4 . $\square$

Corollary 6.3 (All sufficiently large orders). *There is a connected planar graph of treewidth two and cycle rank two with domination root -4 at order 30 and at every order $n \geq 32$.*

Proof. Besides attaching an isolated edge as above, we may attach a disjoint star $K_{1,2}$ by its center to a support vertex. Lemma 6.1 gives an additional factor

$$D(K_{1,2}, x) = x(1 + x)^2 + x^2 = x(x^2 + 3x + 1).$$

At -4 this factor is $-20 \neq 0$; it adds three vertices and a pendant tree. Every integer $n - 30 \geq 2$ is a nonnegative integer combination of two and three. Using one three-vertex attachment for odd $n - 30$, and otherwise only two-vertex attachments, proves the claim. This construction asserts nothing about order 31. $\square$

Theorem 6.4 (Unbounded root multiplicity in connected graphs). *For every integer $k \geq 1$, there is a connected planar graph H_k of order $30k$, treewidth two, and maximum degree seven in which -4 has multiplicity exactly k .*

Proof. Take k disjoint copies of G_{30} , and connect their copies of vertex 10 in a path. Every connecting edge joins support vertices retaining private leaves. Thus

$$D(H_k, x) = D(G_{30}, x)^k.$$

The root has multiplicity exactly k by (5.3). The connecting edges are bridges, so they preserve planarity and do not increase treewidth beyond two. Internal copies of vertex 10 have degree four after the operation; the maximum degree is still seven. $\square$

These results strengthen the bare infinite-family question in [1, Section 4, question 3]: disconnected examples would follow immediately from disjoint union, whereas the constructions here remain connected and structurally sparse.

7 How the search was made finite and reproducible

The successful search was not a uniform enumeration of 30-vertex graphs. It used the sparse core to turn a combinatorial search into rational arithmetic on a catalogue of small rooted trees.

7.1 The catalogue and the exact target equation

A rooted tree is stored as the canonically ordered tuple of the shapes of its children. The empty tuple is a leaf. Recursively enumerate multisets of previously generated trees whose total order is $n - 1$, then place a new root above them. This generates each unlabeled rooted tree of order n exactly once: deleting the root yields a uniquely determined multiset of rooted child components.

For each tree compute $(A(-4), B(-4), C(-4))$ by Lemma 3.2. Retain its rational ratio

$$q = \frac{B(-4)}{A(-4) + B(-4)} \quad \text{when} \quad C(-4) = 0, \quad A(-4) + B(-4) \neq 0.$$

The condition $C(-4) = 0$ is sufficient for a weighted search at the single point -4 ; it need not hold identically for every candidate. All final gadgets happen to satisfy $C(x) = 0$ identically, which makes the polynomial proof particularly clean.

At a cycle site, disjoint branches give a product of their ratios. Store a least-order realization for every attainable product, including the empty forest with ratio one and order zero. Discarding larger realizations of the same ratio is valid for this search: they have identical normalized effects at -4 , and cannot improve its order objective. The catalogue contains all such minimum-cost products within the prescribed forest-size budget.

For a shared star hub with $m \geq 1$ leaves,

$$r_m = \frac{(-4)^m}{-4(-3)^m + (-4)^m} = \frac{4^{m-1}}{4^{m-1} - 3^m}. \quad (7.1)$$

The denominator is nonzero, since a power of four cannot equal a positive power of three. Leaving vertex 0 undecorated reduces the root search, by Proposition 4.1, to

$$c = -21 - r_m(a + b). \quad (7.2)$$

Thus enumerate two stored ratios a, b , compute the required c , and look it up exactly. No tolerance or root approximation occurs. If the three forests have sizes n_a, n_b, n_c , the total order is

$$n = 5 + m + n_a + n_b + n_c. \quad (7.3)$$

The hit has $(m, n_a, n_b, n_c) = (5, 2, 13, 5)$, giving $n = 30$.

7.2 The supplied run and its limitations

The portable reconstruction program uses rooted trees and site forests of order at most 14 and hub sizes $1 \leq m \leq 15$. It starts with the strict target $n < 33$, tries the five-leaf hub first, and then tests the other permitted hub sizes. It retains improving hits and checks the graph of every hit with the separate 2-core polynomial routine. A representative completed run gives:

Quantity	Recorded value
Distinct single-tree ratios	4 198
Distinct forest ratios, including the empty forest	6 118
Exact rational candidate lookups	2 547 880
Best order returned	30
Time in the execution environment	6.154 seconds

The time is an observation on this environment, not a complexity bound or a cross-machine promise. The deterministic counts and the hit are stored in `results/search.json`. The number of rooted trees at each order from 1 through 14 is also recorded there.

In this bounded search, no smaller realization was returned. That is *not* an exhaustive lower bound for all graphs, all bicyclic graphs, or even all rooted attachments to this core: an individual site forest was limited to 14 vertices. Searching other cores, allowing a decorated fourth corner, or allowing larger individual forests can escape the tested class. The abstract minimum remains open in this note.

7.3 Verification and computational trust

The complete coefficient list was recomputed by all three routes in Section 5: the symbolic four-cycle certificate, the 2-core state algorithm, and the independent leaf-subset identity. The latter two are genuinely different enumerations, although all are implemented in the same Python runtime. They are not independent proof-assistant formalizations.

The two general graph algorithms were also compared against direct enumeration of all vertex subsets for *every labeled simple graph on zero through five vertices*: 1100 graphs in total. Further checks used 120 deterministically seeded graphs on six through nine vertices. Five instances of the connected extension in Theorem 6.2 were checked as polynomial identities. All tests passed.

The verification run recorded about 0.23 seconds. The important properties are the exact outputs, not the timing. All polynomial arithmetic is integer arithmetic, and all search ratios use Python's `Fractions.Fraction`. No symbolic algebra package, network service, cached pickle, numerical optimizer, or proprietary system is required to run the supplied code.

8 Conclusions and remaining targets

The attack gives a concrete answer to the selected question: a graph with domination root -4 need not have 33 vertices. The explicit order-30 example also demonstrates that cycle rank two and treewidth two already suffice. The cancellation is additive in three rooted ratios and a single shared-star ratio; it does not require a large or highly connected core.

The demonstrated improvement is an *upper bound*: the minimum order is at most 30. A serious lower-bound project should canonicalize possible small cores and carry exact rooted-state catalogues with a global vertex budget, rather than infer minimality from an unsuccessful restricted search. An especially natural next target is whether some graph on at most 29 vertices has root -4 .

For other integer targets z , the same core identity yields

$$c + r(a + b) = z^3 + 4z^2 + 6z + 3,$$

with the rooted states evaluated at z . This provides a concrete search equation, but this note establishes no graph with root -6 or -8 . Nor does it make a claim that all negative even integers occur. The value of the present result is the fully specified, independently checked counterexample and its complete elementary proof.

A Reproducing the artifact

The archive contains the $\text{\LaTeX}$ source and compiled PDF, a machine-readable edge list and coefficient table, the 118-term leaf certificate, the exact verification report, and the reproducible search program. From the archive's top directory, run

```
python3 code/verify.py
python3 code/search.py
```

Python 3.10 or later is sufficient. The first command recomputes all certificates, runs the tests, and writes `results/verification.json`. The second reconstructs the rooted catalogue from scratch and writes `results/search.json`. Both commands stop with an error if their explicit verification checks fail; the tests do not rely on Python's optionally disabled `assert` statement.

To rebuild the document using a standard TeX Live installation:

```
pdflatex -interaction=nonstopmode -halt-on-error domination_root_30.tex
pdflatex -interaction=nonstopmode -halt-on-error domination_root_30.tex
```

The graph drawing is embedded as TikZ code in this source. No external image or font file is needed. The output PDF was rendered and visually inspected before packaging.

Authorship and status. This note was produced in response to a request for an original research attempt. The construction, proof, and verification artifacts were developed in the session with ChatGPT. The note is not peer reviewed and has not been formally verified in Lean. The exact finite certificate is intended to make external checking straightforward.

References

- [1] S. Alikhani and M. Griswold, *On the Integer Domination Root Conjecture*, arXiv:2608.00109v1, 2026. arxiv.org/abs/2608.00109v1. The selected minimality question is Section 4, question 2; the infinite-family question is question 3. Accessed 18 September 2026.
- [2] M. R. Oboudi, *On the roots of domination polynomial of graphs*, Discrete Applied Mathematics **205** (2016), 126–131. doi:[10.1016/j.dam.2015.12.010](https://doi.org/10.1016/j.dam.2015.12.010). Theorem 7 supplies the trees/unicyclic exclusion used in (2.2); Lemma 1 is the general domination inclusion–exclusion identity.